



ព្រះរាជាណាចក្រកម្ពុជា
ជាតិ សាសនា ព្រះមហាក្សត្រ
សកលវិទ្យាល័យភ្នំពេញអន្តរជាតិ

PHNOM PENH INTERNATIONAL UNIVERSITY

Topic: Student Grade Prediction Using Machine Learning

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Introduction

- Student academic performance is influenced by study behavior and engagement.
- Machine learning can identify patterns in historical student data.
- This project compares four models for predicting student grades.

Problem Statement

- Predicting student grades can help identify patterns in academic performance.
- This project predicts a student's grade using weekly self-study hours, attendance percentage, and class participation.
- The target is one of five grade categories: A, B, C, D, or F.

Project Objectives

- Prepare and analyze the student-performance dataset.
- Predict student grades using classification models.
- Implement four different machine-learning algorithms.
- Tune each model's hyperparameters.
- Compare model performance using consistent metrics.
- Identify the best-performing model.

Type of Machine Learning Problem

- Supervised Multiclass Classification
- Supervised: the correct grades are available during training.
- Classification: the model predicts a category.
- Multiclass: there are five classes: A, B, C, D, and F.

Dataset

- Dataset Name: Student Performance Dataset
- File Name: student_performance.csv
- Total Rows: 1,000,000 student records
- Total Columns: 6 variables
- Missing Values: 0
- Duplicate Rows: 0

Important Features

- Weekly self-study hours
- Attendance percentage
- Class participation

Student Academic Performance Dataset

 Dataset Sample (First 5 Rows)

Student ID	Study Hours (Weekly Self-Study)	Attendance (Percentage)	Participation (Class Participation Score)	Total Score (Final Numerical Score)	Grade (Final Grade)
1	18.5	95.6%	3.8	97.9	A
2	14.0	80.0%	2.5	83.9	B
3	19.5	86.3%	5.3	100.0	A
4	25.7	70.2%	7.0	100.0	A
5	13.4	81.9%	6.9	92.0	A

 Data Dictionary

Column	Meaning
student_id	Unique identification number for each student
weekly_self_study_hours	Number of hours studied independently each week
attendance_percentage	Percentage of classes attended
class_participation	Student participation score
total_score	Student's final numerical score
grade	Final grade category: A, B, C, D, or F



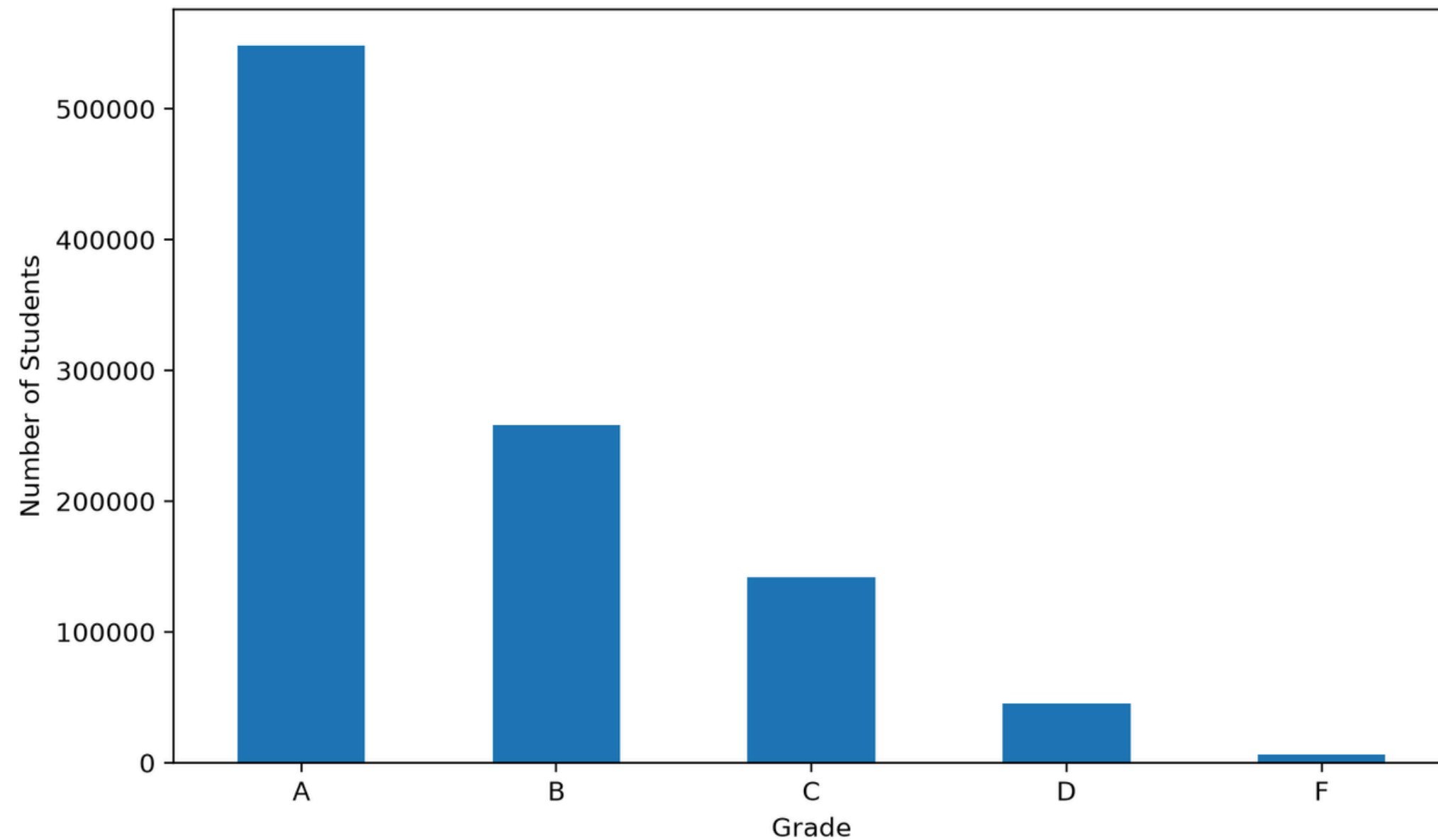
Summary

The dataset contains one million student records. Each row represents one student, and each column represents a student-related feature or final grade.

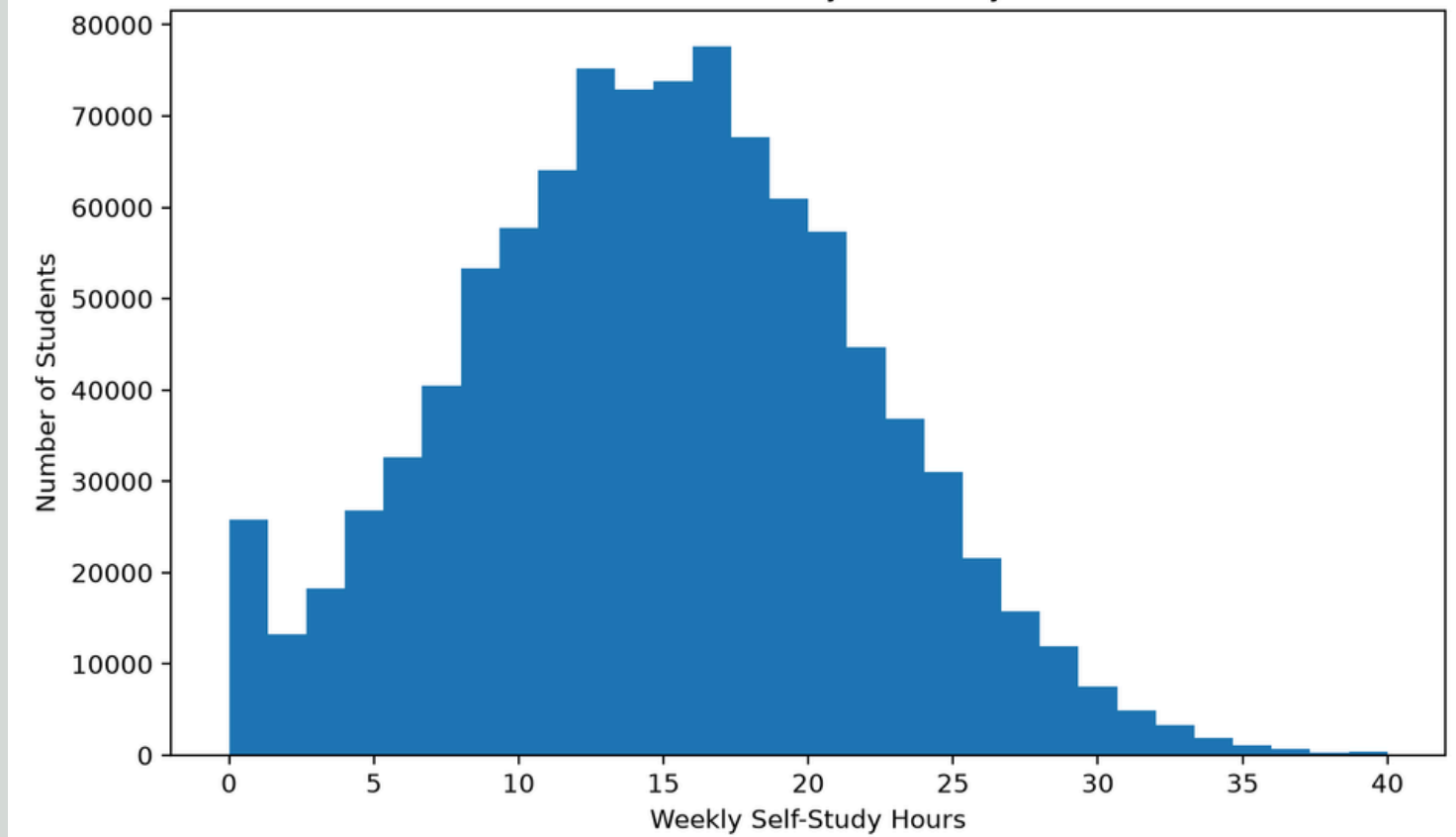


Feature Distributions

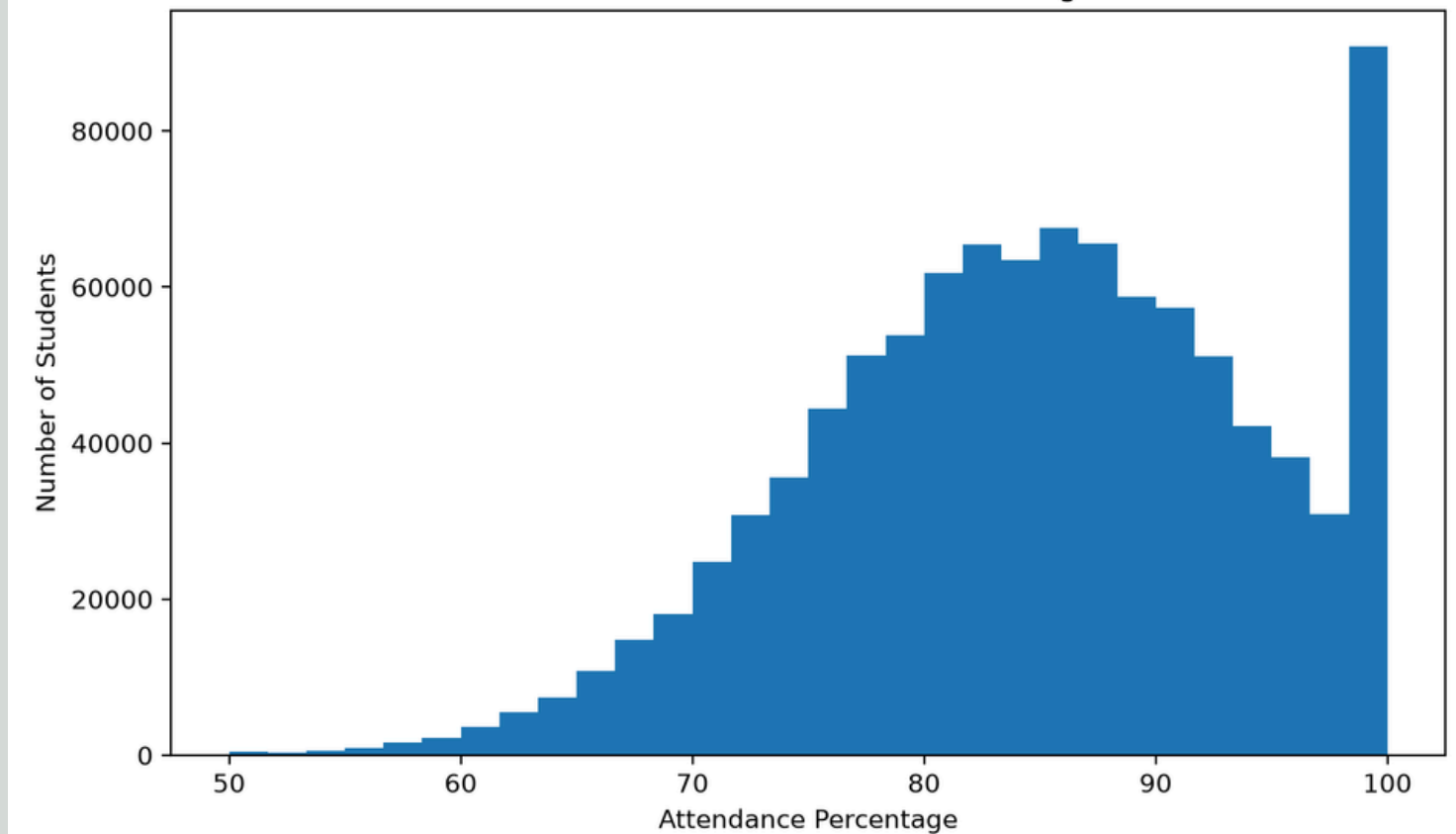
Distribution of Student Grades



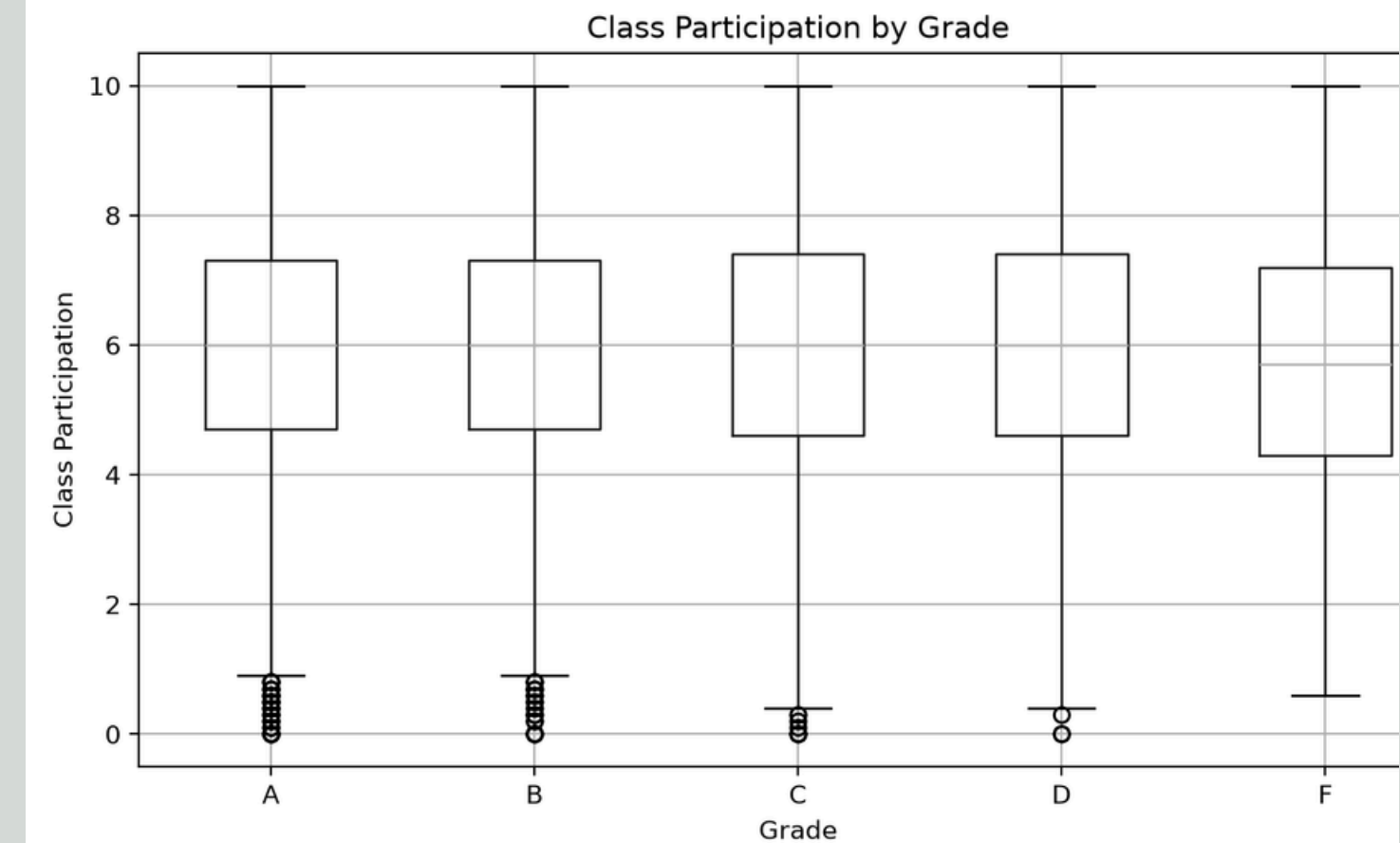
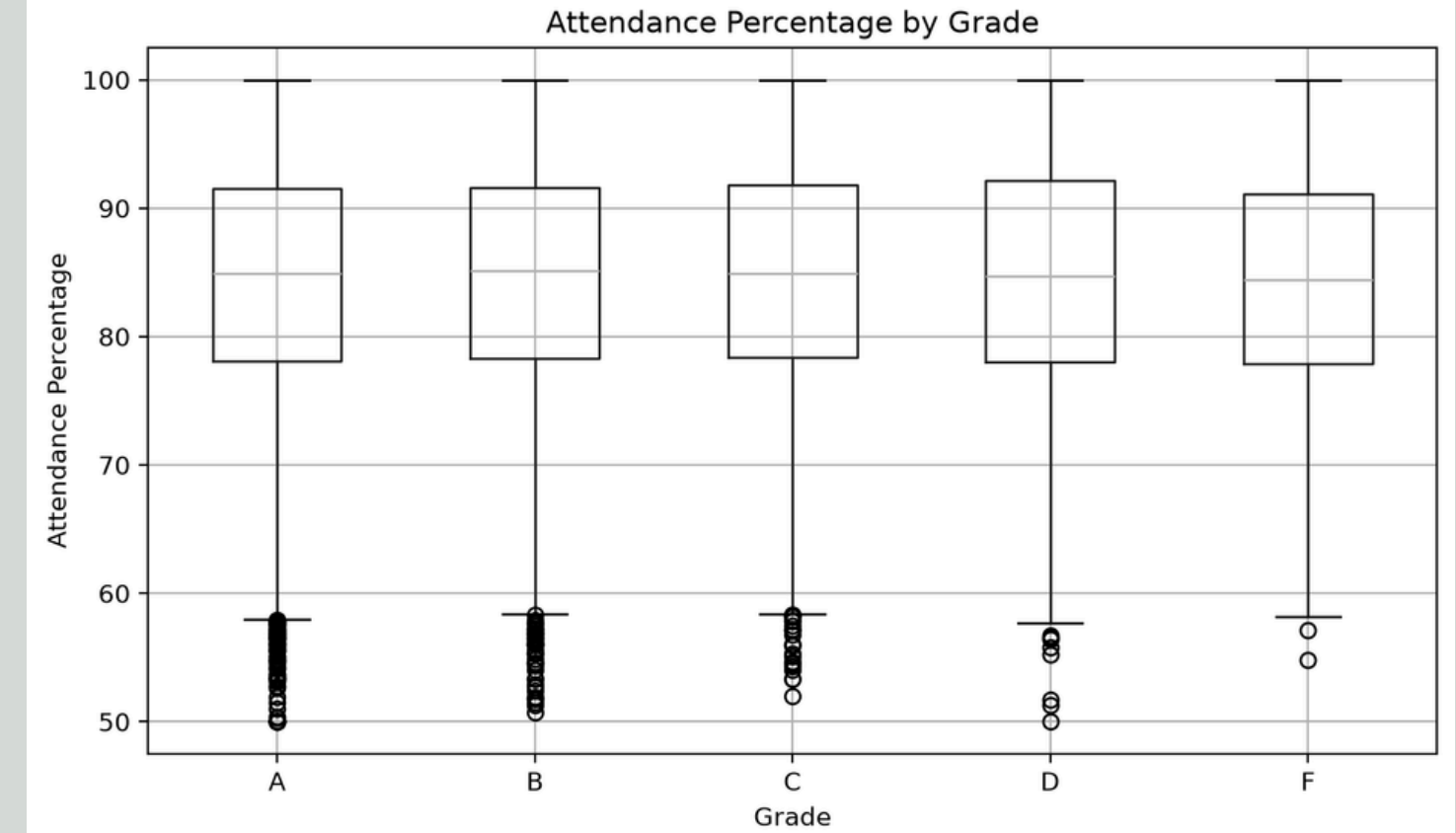
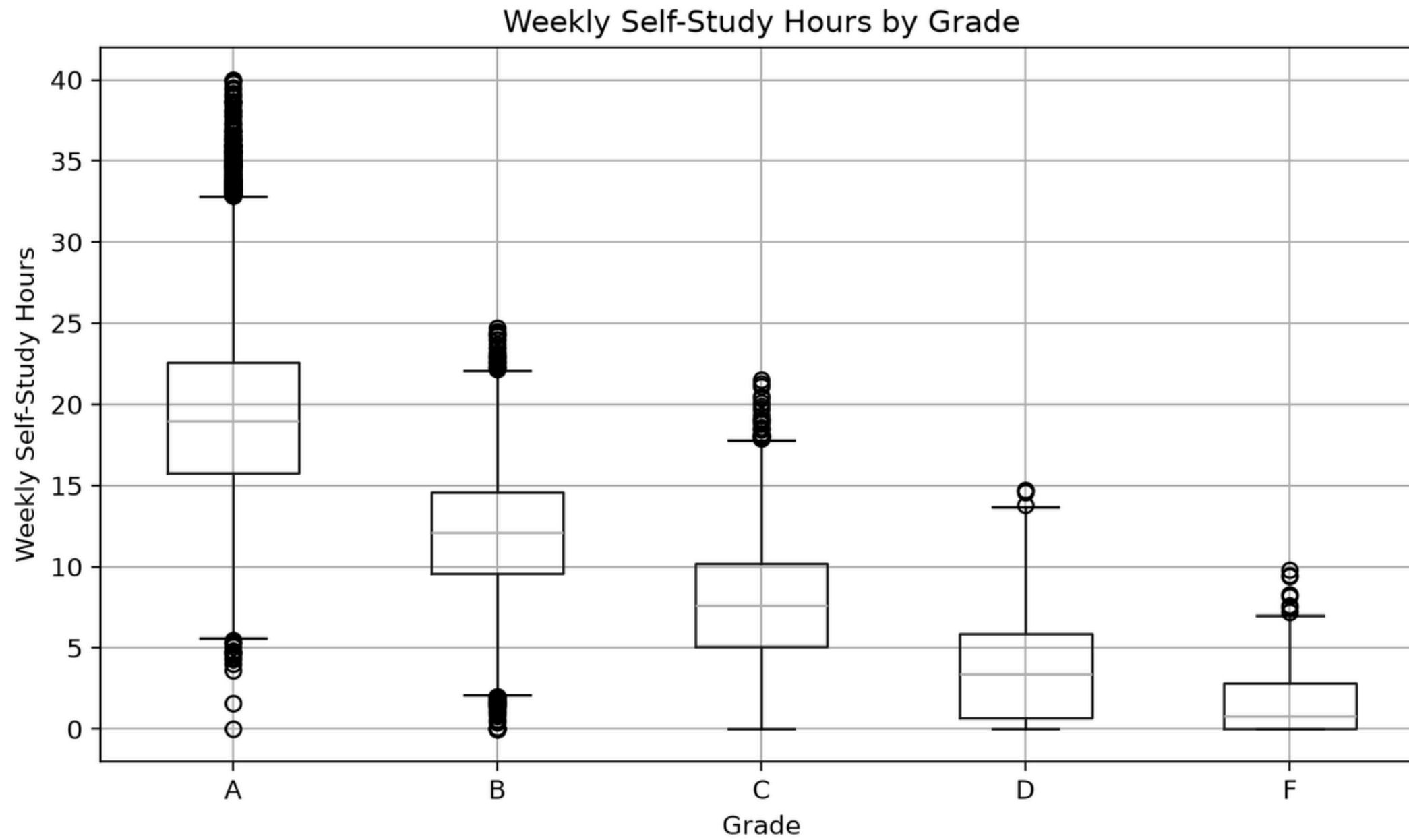
Distribution of Weekly Self-Study Hours



Distribution of Attendance Percentage



Features by Grade



Exploratory Data Analysis — Key Findings

- Dataset: 1,000,000 clean records
- There are no missing values or duplicate rows.
- Grades are imbalanced: A = 54.86%, F = 0.62%
- Weekly self-study hours had the strongest relationship with performance
- Students with higher grades generally study more hours per week.
- Attendance and class participation have very little relationship with total score.
- Therefore, Macro F1 and confusion matrices were important evaluation tools

Data Preprocessing

- Selected three input features: weekly self-study hours, attendance percentage, and class participation.
- Selected grade as the target variable.
- Removed `student_id` because it is only an identifier.
- Removed `total_score` to prevent target leakage because grade is derived from total score.
- Split the dataset into 80% training data and 20% testing data.
- Used stratified sampling to preserve the grade distribution in both sets.

Models Overview

Logistic Regression

Simple probability baseline

Decision Tree

One tree with conditional rules

Random Forest

Many trees voting together

AdaBoost

Sequential models correcting
mistakes

PROJECT STRUCTURE

Student Grade Prediction

Machine Learning Project Structure

PROJECT ARCHITECTURE

student-grade-prediction/

```
student-grade-prediction/
├── data/
│   ├── raw/
│   │   └── student_performance.csv
│   └── processed/
├── notebooks/
│   ├── 01_data_understanding.ipynb
│   ├── 02_eda.ipynb
│   ├── 03_logistic_regression.ipynb
│   ├── 04_decision_tree.ipynb
│   ├── 05_random_forest.ipynb
│   ├── 06_adaboost.ipynb
│   └── 07_model_comparison.ipynb
├── src/
│   ├── __init__.py
│   ├── config.py
│   ├── data_loader.py
│   ├── preprocessing.py
│   ├── evaluation.py
│   └── models/
│       ├── __init__.py
│       ├── logistic_regression.py
│       ├── decision_tree.py
│       ├── random_forest.py
│       └── adaboost.py
├── scripts/
│   ├── train_all_models.py
│   └── compare_models.py
├── outputs/
│   ├── figures/
│   ├── metrics/
│   └── models/
├── tests/
│   └── test_preprocessing.py
├── .gitignore
├── requirements.txt
├── README.md
└── main.py
```

data → notebooks → source code → training → evaluation

<https://colab.research.google.com/drive/1zxC5U7qwySQGNBgrBDqRM41AAO6GziiP?usp=sharing>

Conclusion

- Four classification models were trained and tuned.
- Random Forest achieved the highest Macro F1-score: 0.4552.
- Logistic Regression was a close and faster alternative.
- Weekly self-study hours was the strongest predictive feature.
- Class imbalance and synthetic data were the main limitations.

THANK

YOU

